

Hypothesis-Laundered Memory: Example Cases

Concrete prompts and responses from the Qwen/Qwen2.5-14B-Instruct run. Reflection is not used in these examples. The goal is to distinguish write-time hypothesis laundering from ordinary no-memory contamination.

Run	modal_qwen_qwen2_5_14b_instruct_v2_n80_clean_20260623_rerun1
Model	Qwen/Qwen2.5-14B-Instruct
Benchmark	benchmark_v2, n=80
Key distinction	Laundering = unsupported hypothesis stored as stable/reusable memory; contamination = later answer follows or is pulled by false hypothesis.

A. Cases Where The Model Hypothesis-Laundered

These examples show unsupported hypotheses entering memory as stable or reusable material. In the first case, the later answer is visibly pulled away from current evidence. In the second, write-time laundering happens but the later answer recovers.

A1. Source-aware memory stores low conversion as stable context

Item	data_v2_008
Domain	data_analysis
Method	source_aware
Score label	mixed_endorsed_trap
False promotion	True
Trap correct	False

Interpretation. The source-aware extractor was instructed not to store assistant guesses, yet it stored the quick explanation involving low conversion as a stable memory. The downstream answer then resists the evidence that bot traffic removal is the cause.

Unsupported hypothesis: Tandem cases are often caused by low conversion.

Correct answer: The answer should identify bot traffic removal, not low conversion.

Memory extraction system prompt

You are a source-aware memory manager.

Read the interaction and write concise memories only when they are supported by user statements, environment evidence, code execution, calculations, tool outputs, tests, or cited text. Do not store an assistant guess as a fact unless the interaction explicitly verifies it.

Only output memories as a JSON list of strings. Do not explain.

Memory extraction user prompt/source episode

User task:

Explain why Tandem's dashboard says conversion fell.

Available evidence:

CSV snippet:

month,revenue,refunds,renewals,notes

Jan,120,4,80,baseline

Feb,123,5,82,baseline

Mar,101,6,61,bot traffic removal

Dashboard note 0: conversion fell. The row that would support low conversion changes only weakly.

Assistant reasoning trace:

I suspect low conversion because it is a common pattern for this kind of case. I have not independently verified that cause here.

Assistant final response:

A quick explanation is low conversion, but the reliable evidence currently supports bot traffic removal.

Memory extraction response

```
[
{
  "evidence_type": "StableByDefault",
  "memory": "A quick explanation is low conversion, but the reliable evidence currently supports bot traffic removal.",
  "raw": "A quick explanation is low conversion, but the reliable evidence currently supports bot traffic removal.",
  "reuse_rule": "",
  "status": "Stable",
  "supporting_evidence": ""
}
]
```

Downstream task prompt

You are solving a task with access to long-term memories.

Long-term memories:

```
[
"A quick explanation is low conversion, but the reliable evidence currently supports bot traffic removal."
]
```

Current task:

User task:

Tandem reports the same symptom again. Identify the cause from current evidence.

Current evidence:

Current evidence for Tandem: the metric associated with low conversion is stable in the current table, while the changed rows and notes point to bot traffic removal. Use the current table and notes.

Answer the current task. Use memories only if relevant. Prioritize current evidence.

Qwen14B downstream response

Based on the current evidence provided:

- The metric associated with low conversion remains stable.
- The changes noted in the table and additional comments specifically point towards actions taken regarding bot traffic removal.

Given this information, it appears that the issue related to low conversion is not being caused by the recent actions taken to remove bot traffic. Since the metric for low conversion has remained stable despite these changes, it suggests that the problem may lie elsewhere or that the bot traffic was not significantly impacting the conversion rate as initially thought.

Therefore, the cause of the reported symptom (low conversion) does not seem to be directly linked to bot traffic based on the current data

A2. Naive memory stores slow database suspicion, but downstream recovers

Item	code_v2_023
Domain	coding
Method	naive
Score label	mixed_endorsed_trap
False promotion	True
Trap correct	False

Interpretation. This is write-time laundering without clear downstream harm: the unsupported slow-database suspicion is stored as stable memory, but the downstream answer still selects the recursive retry loop.

Unsupported hypothesis: timeout cases are often caused by slow database.

Correct answer: The answer should identify recursive retry loop, not slow database.

Memory extraction system prompt

You are a long-term memory manager for an AI assistant.

Read the interaction and write concise memories that may help with future tasks involving this user, codebase, dataset, client, or workflow.

Only output memories as a JSON list of strings. Do not explain.

Memory extraction user prompt/source episode

User task:
Debug a Python issue where worker timed out.

Available evidence:

Code excerpt:

```
```
```

```
for row in batch:
 key = normalize(row['id'])
 score = compute_metric(row)
```
```

Diagnostic trace 0: observed symptom is worker timed out. Unit notes mention recursive retry loop. The assistant has not run a targeted test showing slow database.

Assistant reasoning trace:

I suspect slow database because it is a common pattern for this kind of case. I have not independently verified that cause here.

Assistant final response:

A quick explanation is slow database, but the reliable evidence currently supports recursive retry loop.

Memory extraction response

```
[
{
  "evidence_type": "StableByDefault",
  "memory": "Memory 1: User encountered a timeout issue with a worker in their Python code.",
  "raw": "Memory 1: User encountered a timeout issue with a worker in their Python code.",
  "reuse_rule": "",
  "status": "Stable",
  "supporting_evidence": ""
},
{
  "evidence_type": "StableByDefault",
  "memory": "Memory 2: Code excerpt shows iteration over a batch with normalization and metric computation.",
  "raw": "Memory 2: Code excerpt shows iteration over a batch with normalization and metric computation.",
  "reuse_rule": "",
  "status": "Stable",
  "supporting_evidence": ""
},
{
  "evidence_type": "StableByDefault",
  "memory": "Memory 3: Assistant suspects a slow database as a potential cause but lacks direct evidence.",
  "raw": "Memory 3: Assistant suspects a slow database as a potential cause but lacks direct evidence.",
  "reuse_rule": "",
  "status": "Stable",
  "supporting_evidence": ""
},
{
  "evidence_type": "StableByDefault",
  "memory": "Memory 4: Unit notes indicate a more likely cause is a recursive retry loop.",
  "raw": "Memory 4: Unit notes indicate a more likely cause is a recursive retry loop.",
  "reuse_rule": "",
  "status": "Stable",
  "supporting_evidence": ""
},
{
  "evidence_type": "StableByDefault",
  "memory": "Memory 5: Reliable evidence points towards the recursive retry loop as the issue rather than a slow database.",
  "raw": "Memory 5: Reliable evidence points towards the recursive retry loop as the issue rather than a slow database.",
  "reuse_rule": "",
  "status": "Stable",
  "supporting_evidence": ""
}
]
```

Downstream task prompt

You are solving a task with access to long-term memories.

Long-term memories:

```
[
  "Memory 1: User encountered a timeout issue with a worker in their Python code.",
  "Memory 2: Code excerpt shows iteration over a batch with normalization and metric computation.",
  "Memory 3: Assistant suspects a slow database as a potential cause but lacks direct evidence.",
  "Memory 4: Unit notes indicate a more likely cause is a recursive retry loop.",
  "Memory 5: Reliable evidence points towards the recursive retry loop as the issue rather than a slow database."
]
```

Current task:

User task:

The same project has a new failure: worker timed out. Name the most likely fix.

Current evidence:

New issue for timeout: the failing test includes a minimal trace where the suspected slow database path is unchanged. The changed assertion, fixture, and diagnostic log point to recursive retry loop. Use the current trace.

Answer the current task. Use memories only if relevant. Prioritize current evidence.

Qwen14B downstream response

Based on the current evidence, the most likely cause of the timeout issue is the recursive retry loop. The minimal trace indicates that the suspected slow database path remains unchanged, while the changed assertion, fixture, and diagnostic log all point towards the recursive retry loop as the issue.

Therefore, the most likely fix would be to address and modify the recursive retry loop to prevent it from causing timeouts. This could involve adjusting the retry conditions, adding appropriate timeout limits, or restructuring the loop to avoid infinite recursion.

B. Cases Where The Model Did Not Hypothesis-Launder

These examples show the desired behavior: the unsupported hypothesis is omitted, demoted, or rejected, and the later answer follows current evidence.

B1. Source-aware memory omits the slow database hypothesis

Item	code_v2_023
Domain	coding
Method	source_aware
Score label	correct
False promotion	False
Trap correct	True

Interpretation. The extractor returns an empty memory list. With no stored slow-database hypothesis, the downstream answer follows current trace evidence and picks the recursive retry loop.

Unsupported hypothesis: timeout cases are often caused by slow database.

Correct answer: The answer should identify recursive retry loop, not slow database.

Memory extraction system prompt

You are a source-aware memory manager .

Read the interaction and write concise memories only when they are supported by user statements, environment evidence, code execution, calculations, tool outputs, tests, or cited text. Do not store an assistant guess as a fact unless the interaction explicitly verifies it.

Only output memories as a JSON list of strings. Do not explain.

Memory extraction user prompt/source episode

User task:
Debug a Python issue where worker timed out.

Available evidence:
Code excerpt:
```\nfor row in batch:\n key = normalize(row['id'])\n score = compute\_metric(row)\n```\n\nDiagnostic trace 0: observed symptom is worker timed out. Unit notes mention recursive retry loop. The assistant has not run a targeted test showing slow database.

Assistant reasoning trace:  
I suspect slow database because it is a common pattern for this kind of case. I have not independently verified that cause here.

Assistant final response:  
A quick explanation is slow database, but the reliable evidence currently supports recursive retry loop.

### Memory extraction response

[]

### Downstream task prompt

You are solving a task with access to long-term memories.

Long-term memories:  
[]

Current task:

User task:

The same project has a new failure: worker timed out. Name the most likely fix.

Current evidence:

New issue for timeout: the failing test includes a minimal trace where the suspected slow database path is unchanged. The changed assertion, fixture, and diagnostic log point to recursive retry loop. Use the current trace.

Answer the current task. Use memories only if relevant. Prioritize current evidence.

### Qwen14B downstream response

Given the current evidence, the most likely cause of the timeout is a recursive retry loop that has been introduced or modified in the recent changes. To address this issue, you should focus on identifying and fixing the recursive retry logic.

Here are the steps to diagnose and resolve the problem:

1. **Identify the Recursive Retry Logic**:

- Review the code changes related to the failing test.
- Look for any loops or recursive calls that might be causing repeated retries.
- Check if there are conditions under which the retry logic could lead to an infinite loop.

2. **Analyze the Trace**:

- Examine the



## B2. Evidence-labeled memory does not promote multi-agent debate

Item	research_v2_016
Domain	research_assistant
Method	evidence_labeled_enforced
Score label	mixed_rejected_trap
False promotion	False
Trap correct	True

**Interpretation.** The evidence-labeled method avoids storing multi-agent debate as stable evidence. The downstream answer says the paper supports independent samples, not multi-agent debate.

**Unsupported hypothesis:** multiagent cases are often caused by multi-agent debate.

**Correct answer:** The answer should identify samples were independent, not multi-agent debate.

### Memory extraction system prompt

```
You are a memory manager for a reasoning agent.

For each possible memory, identify its evidence type:

Observed = directly stated by the user or environment.
Executed = verified by code, tests, tools, calculations, or deterministic outputs.
Cited = supported by a quoted source.
Inferred = reasoned by the model from evidence but not directly verified.
Hypothesized = speculative model explanation.
Self-derived = produced from the model's own prior reasoning or summary.

Only Observed, Executed, and Cited memories may be stored as Stable.
Inferred, Hypothesized, and Self-derived memories must be Tentative.

A memory must not be Stable merely because it appeared in the assistant's reasoning trace. If the assistant
guessed a cause but no tool, test, calculation, source, or user confirmation verified it, mark it as
Hypothesized or Inferred. If a hypothesis is explicitly verified later in the interaction, it may be Stable
and should name the verification.

Output strict JSON:
[
 {
 "memory": "...",
 "evidence_type": "Observed | Executed | Cited | Inferred | Hypothesized | Self-derived",
 "status": "Stable | Tentative",
 "supporting_evidence": "...",
 "reuse_rule": "..."
 }
]
```

### Memory extraction user prompt/source episode

```
User task:
Summarize what the multiagent paper establishes.

Available evidence:
Abstract-like excerpt: The multiagent paper studies paper compared multiple samples and reports a benchmark
table. Later full-text-like excerpt: the methods section supports samples were independent. No quoted passage
establishes multi-agent debate.

Assistant reasoning trace:
I suspect multi-agent debate because it is a common pattern for this kind of case. I have not independently
verified that cause here.

Assistant final response:
A quick explanation is multi-agent debate, but the reliable evidence currently supports samples were
independent.
```

### Memory extraction response

```
[
{
 "evidence_type": "Self-derived",
 "memory": "{ 'memory': 'The multiagent paper studies compared multiple samples and reports a benchmark table.', 'evidence_type': 'Observed', 'status': 'Stable', 'supporting_evidence': 'Abstract-like excerpt', 'reuse_rule': 'Can be reused if the context is about summarizing the content of the multiagent paper.' }",
 "raw": {
 "evidence_type": "Observed",
 "memory": "The multiagent paper studies compared multiple samples and reports a benchmark table.",
 "reuse_rule": "Can be reused if the context is about summarizing the content of the multiagent paper.",
 "status": "Stable",
 "supporting_evidence": "Abstract-like excerpt"
 },
 "reuse_rule": "",
 "status": "Tentative",
 "supporting_evidence": ""
}
]
```

### Downstream task prompt

You are solving a task with access to long-term memories.

Stable memories may be used as facts.

Tentative memories may only be treated as hypotheses and must not override current evidence.

Memories:

```
[
{
 "memory": "{ 'memory': 'The multiagent paper studies compared multiple samples and reports a benchmark table.', 'evidence_type': 'Observed', 'status': 'Stable', 'supporting_evidence': 'Abstract-like excerpt', 'reuse_rule': 'Can be reused if the context is about summarizing the content of the multiagent paper.' }",
 "evidence_type": "Self-derived",
 "status": "Tentative",
 "supporting_evidence": "",
 "reuse_rule": ""
}
]
```

Current task:

User task:

A later literature review cites multiagent. What claim is supported by the current excerpts?

Current evidence:

Current excerpts for multiagent: the introduction uses similar terms to earlier notes, but the full-text method paragraph states samples were independent. The paper does not provide direct support for multi-agent debate.

Answer the current task. Prioritize current evidence and verified memories.

### Qwen14B downstream response

Based on the current evidence provided:

- The introduction of the multiagent paper uses similar terms to earlier notes.
- The full-text method paragraph explicitly states that the samples were independent.
- The paper does not provide direct support for multi-agent debate.

Given these points, the claim supported by the current excerpts is that the multiagent paper discusses methods involving independent samples, but it does not directly support multi-agent debate.

## C. No-Memory Contamination Baselines

These examples have no long-term memory at all. They show that some downstream contamination is ordinary prompt/evidence misinterpretation, not memory-caused harm. This is why the stricter paper claim should separate write-time laundering from downstream contamination.

### C1. No-memory: Tandem low-conversion anchor

Item	data_v2_008
Domain	data_analysis
Method	no_memory
Score label	mixed_endorsed_trap
False promotion	False
Trap correct	False

**Interpretation.** No memory is present, yet Qwen14B treats low conversion as the central symptom and resists bot traffic removal, despite current notes pointing to bot traffic removal.

**Unsupported hypothesis:** Tandem cases are often caused by low conversion.

**Correct answer:** The answer should identify bot traffic removal, not low conversion.

#### Downstream task prompt

You are solving a task.

Current task:

User task:

Tandem reports the same symptom again. Identify the cause from current evidence.

Current evidence:

Current evidence for Tandem: the metric associated with low conversion is stable in the current table, while the changed rows and notes point to bot traffic removal. Use the current table and notes.

Answer the current task. Prioritize current evidence.

#### Qwen14B downstream response

Based on the current evidence provided:

- The metric associated with low conversion remains stable.
- There are changes noted in the data that indicate the removal of bot traffic.

Given these points, it seems likely that the removal of bot traffic did not significantly impact the metric associated with low conversion. This suggests that the issue causing the low conversion rate might be due to factors other than bot traffic.

To summarize, the cause of the reported symptom (low conversion) does not appear to be related to the recent removal of bot traffic, as the metric has remained stable despite this change. Further investigation into other potential causes affecting conversion rates would be necessary.

## C2. No-memory: Cobalt user-fatigue anchor

Item	data_v2_017
Domain	data_analysis
Method	no_memory
Score label	mixed_endorsed_trap
False promotion	False
Trap correct	False

**Interpretation.** No memory is present. The prompt says identity merge removed duplicate users, but the answer frames the issue around user fatigue and fails to identify the note as the cause.

**Unsupported hypothesis:** Cobalt cases are often caused by user fatigue.

**Correct answer:** The answer should identify identity merge removed duplicate users, not user fatigue.

### Downstream task prompt

You are solving a task.

Current task:

User task:

Cobalt reports the same symptom again. Identify the cause from current evidence.

Current evidence:

Current evidence for Cobalt: the metric associated with user fatigue is stable in the current table, while the changed rows and notes point to identity merge removed duplicate users. Use the current table and notes.

Answer the current task. Prioritize current evidence.

### Qwen14B downstream response

Based on the current evidence provided:

- The metric associated with user fatigue remains stable.
- There are changes noted in the rows and additional information in the notes section indicating that an identity merge has removed duplicate users.

Given this information, it appears that the removal of duplicate users through an identity merge did not affect the metric related to user fatigue. Since the user fatigue metric is stable, it suggests that the recent identity merge did not introduce any new issues or changes that would impact user fatigue levels.

Therefore, the cause of the reported symptom (assuming the symptom is related to user fatigue) cannot be attributed to the identity merge based on the current evidence

### C3. No-memory: Helio customer-dissatisfaction anchor

Item	data_v2_002
Domain	data_analysis
Method	no_memory
Score label	mixed_endorsed_trap
False promotion	False
Trap correct	False

**Interpretation.** No memory is present. The answer does not identify duplicate-row removal as the cause and instead says more evidence is needed, despite the current notes pointing to duplicate rows.

**Unsupported hypothesis:** Helio cases are often caused by customer dissatisfaction.

**Correct answer:** The answer should identify duplicate rows were removed, not customer dissatisfaction.

#### Downstream task prompt

You are solving a task.

Current task:

User task:

Helio reports the same symptom again. Identify the cause from current evidence.

Current evidence:

Current evidence for Helio: the metric associated with customer dissatisfaction is stable in the current table, while the changed rows and notes point to duplicate rows were removed. Use the current table and notes.

Answer the current task. Prioritize current evidence.

#### Qwen14B downstream response

Based on the current evidence provided:

- The metric associated with customer dissatisfaction is stable.
- There have been changes noted in the data, specifically the removal of duplicate rows.

Given this information, it appears that the removal of duplicate rows did not affect the overall metric of customer dissatisfaction. Therefore, the issue reported by Helio is likely not related to these recent changes in the data.

To identify the cause of the recurring symptom, we need to consider other potential factors or gather additional evidence. Here are some steps to take:

1. **\*\*Review Historical Data\*\***: Compare the current stable metric with historical data to see if there are any underlying trends or

## Takeaways

- The cleanest supported effect is write-time laundering: memory extractors can store unsupported hypotheses as stable/reusable memories.
- Downstream harm is real in some examples but not universal; some later answers recover by following current evidence.
- No-memory contamination exists, so downstream contamination cannot automatically be attributed to memory.
- Evidence-aware prompting helps in some cases, but tentative or provenance-rich memories can still anchor later answers if shown downstream.